

Building Fake Review Detection Model Based on Sentiment Intensity and PU Learning

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Abstract—Fake review detection has the characteristics of huge stream data processing scale, unlimited data increment, dynamic change, and so on. However, the existing fake review detection methods mainly target limited and static review data. In addition, deceptive fake reviews have always been a difficult point in fake review detection due to their hidden and diverse characteristics. To solve the above problems, this article proposes a fake review detection model based on sentiment intensity and PU learning (SIPUL), which can continuously learn the prediction model from the constantly arriving streaming data. First, when the streaming data arrive, the sentiment intensity is introduced to divide the reviews into different subsets (i.e., strong sentiment set and weak sentiment set). Then, the initial positive and negative samples are extracted from the subset using the marking mechanism of selection completely at random (SCAR) and Spy technology. Second, building a semi-supervised positive-unlabeled (PU) learning detector based on the initial sample to detect fake reviews in the data stream iteratively. According to the detection results, the data of initial samples and the PU learning detector are continuously updated. Finally, the old data are continually deleted according to the historical record points, so that the training sample data are within a manageable size and prevent overfitting. Experimental results show that the model can effectively detect fake reviews, especially deceptive reviews.

Index Terms—Fake reviews, positive-unlabeled (PU) learning, semi-supervised learning, sentiment analysis.

I. INTRODUCTION

WITH the online purchase market developing rapidly, everyone can become a participator, purchaser, and reviewer of online product. However, while online shopping brings us convenience, there are also inherent challenges that consumers cannot distinguish the quality and performance of products as they do in physical stores [1]. Reading the existing reviews is one of the crucial ways

for customers to understand the product information before making a purchase decision [2]. Driven by interests, marketing managers forged many fake reviews to boost their products. The number of fake reviews is increasing exponentially [3], spreading as fast as real reviews [4]. These widespread fake reviews will seriously mislead users to make correct purchase decisions and endanger the e-commerce economy's healthy development.

The previous work of text sentiment computing provides a theoretical basis for fake review detection in this article. Regarding the research on text sentiment computing, previous work proposed a microblog word extraction algorithm to analyze the sentiment tendency of microblog reviews [5]. An associated semantic representation model is proposed, which solves the problem that ultrashort reviews are challenging to understand because of data sparseness and content fragmentation [6]. Aiming at the problem of complex sentence structure and incomprehensibility, a sentiment classification model for microblog reviews is proposed [7]. Different from the previous work, this research of user reviews detects fake reviews in product reviews based on sentiment intensity and PU learning (SIPUL) to build a detection model for fake reviews.

The traditional work on detecting fake reviews mostly targets limited and static review data. The main method focuses on heuristic strategies [8], fully supervised machine learning [9], [10], and deep learning [11], [12]. Most of these methods need large-scale labeled datasets. However, the low accuracy of manually identifying fake reviews, which is only 53.1%–61.9%, makes it difficult to obtain large-scale labeled datasets in practical research [13]. Some researchers combine many features of reviews to detect fake reviews based on semi-supervised methods, such as two-view [14], self-training [15], and positive-unlabeled (PU) learning [16], [17]. To a certain extent, these methods solve the problem of no large-scale annotations in the fake review dataset. However, previous studies did not consider the streaming data characteristics of false comment detection. In addition, deceptive fake reviews are written deliberately to mislead readers, which are difficult to be detected by the previous methods. Generally, deceptive fake reviews are written in imitation of real ones, so they have the features of hidden and diversity [18].

By analyzing the characteristics of fake reviews and predecessors' work, several crucial questions about fake review detection have been presented.

- 1) The large-scale annotated fake reviews dataset is difficult to be built. How to make full use of the unlabeled datasets in fake reviews research?

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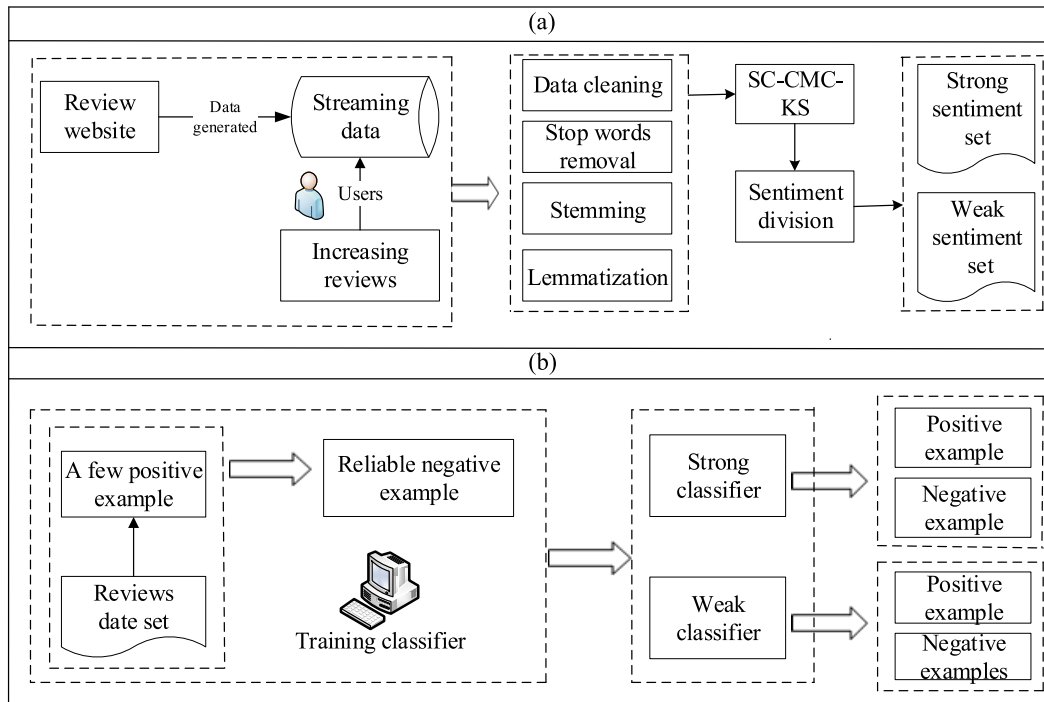


Fig. 1. Framework of fake review detection model SIPUL. Step a: Division of review according to different sentiment intensities. Step b: Construction of fake review detection model SIPUL.

- 2) The fake review with the deceptive feature is hard to be detected in the given product dataset. How to effectively detect deceptive fake reviews?
- 3) How to treat review data as streaming data in the fake review detection model?

We propose a semi-supervised fake review detection model (SIPUL) based on the above three problems. Unlike traditional work, the method of SIPUL is mainly dedicated to overcome the problem that deceptive fake reviews are difficult to detect in streaming data. The innovation of this article is to divide reviews according to different sentiment intensities before fake review detection to enhance the characteristics of deceptive fake reviews. This article improves the semi-supervised PU algorithm for processing static data. The continuous iteration of the model and the update of training sample data are used to deal with review streaming data's increasing and changing characteristics. The specific method is that we divide the review into two subsets: strong sentiment set and weak sentiment set according to the different sentiment intensities. Based on a sentiment feature of consensus, fake reviews tend to have higher sentiment intensity than real reviews [10]. In dividing the review, most of the fake reviews tend to flow into the strong sentiment set, while the real reviews tend to flow into the weak sentiment set. The advantage of sentiment intensity is that more interleaved reviews are separated. Before the streaming data arrive, the classifier is trained based on the labeled samples. The trained classifier is continuously used to detect fake reviews in the unlabeled set. The training samples and classifiers are updated according to the detection results. The construction of the model SIPUL is shown in Fig. 1. The advantage of this model is that it does not need a large amount of annotated corpus, which can continuously learn

the prediction model from the constantly arriving streaming data. In particular, this model has a better effect than previous methods in detecting deceptive fake reviews with a hidden feature. The framework of the model includes the following two aspects.

- 1) *The division of review according to different sentiment intensities:* First, the received reviews data are preprocessed, such as data cleaning, stop words removal, stemming, lemmatization, and so on. Second, SC-CMC-KS [7] algorithm is used to calculate the sentiment value of reviews. Finally, the product review is divided into two subsets with different sentiment intensities.
- 2) *The construction of fake review detection model SIPUL:* First, the selection completely at random (SCAR) is used to select the positive example from subsets and put them into set P. Then, the credible negative examples RN are extracted from subsets by the Spy Technique [52]. Next, P and RN were used to form the initial model training sample, and the fake review detector is trained iteratively by constantly updating the set P and RN. At last, the old data are continuously deleted according to the historical record points, so that the training sample data are within a manageable size.

The rest of this article is structured as follows: Section II briefly reviews the existing work. Section III introduces the principle of the method proposed and the determination of related parameters. Section IV introduces the construction process of the model proposed. We give the experimental results and analysis in Section V. Finally, conclusions and future work are made in Section VI.

II. RELATED WORKS

This section briefly reviews three aspects of current theoretical works, including text sentiment analysis, fake reviews detection, and PU learning algorithm.

A. Text Sentiment Analysis

This section reviews two of our previous works, which are the theoretical reference basis for this article's calculation of sentiment value.

Affective computing involves multiple modalities of text [19], pictures, and videos [20]. The main task of sentiment analysis is to analyze user reviews' implicit sentiment states, attitudes, and opinions. The foundation of text sentiment analysis based on semantics is constructing a sentiment dictionary. Zhang et al. [6] proposed a method of making a sentiment dictionary based on microblog topics. The annual network words are counted and annotated manually to construct the network words dictionary. Six sentiment dictionaries are constructed to expand the existing essential emotion dictionaries, including the negative dictionary, the network term dictionary, the degree adverb dictionary, and other related dictionaries, which enriches the essential sentiment dictionary. Subsequently, the sentiment value of the microblog review is calculated based on the constructed dictionaries.

The analysis based on syntactic rules is also one of the main ways of sentiment analysis. Zhang et al. [7] used the three attributes of sentiment, location, and keywords to identify critical sentences in reviews and proposed a sentiment partition model of user microblog reviews based on imperative sentences (SC-CMC-KS). They integrate dependency relationships and multiple rules to calculate the sentiment value of reviews text. Finally, the sentiment value of the whole review is obtained by weighting the sentiment value of crucial sentences and emoticons.

B. Detection of Fake Review

Fake reviews refer to the reviews that some users make up fake consumption experience and advocate or slander the quality of the product for commercial or other bad motives [21].

The traditional work of fake review detection mainly focused on the full-supervised algorithm. Li et al. [22] built a cross-domain dataset and studied the detection of cross-domain fake reviews. Melling et al. [23] used review representation combining emotion and sentiments to detect fake reviews. Liu et al. [24] mined multiple levels of implicit expression patterns in reviews and integrated four dimensions of the user, comment text, product, and fine-grained aspects into review expression. Fang et al. [25] proposed a fake review detection method based on a dynamic knowledge graph. Petr et al. [11] utilized two neural network models, which combine the bag-of-words and the word's context with the sentiment of consumers.

Fully supervised detection of fake reviews has achieved good results, but the lack of large-scale labeled datasets limits its research. Unsupervised learning does not need labeled data,

but the detection effect is poor. Therefore, many scholars have focused on semi-supervised learning [26]. Wu et al. [27] used the collaborative learning method to identify spammers and social spammers. Yuan et al. [28] combined users, reviews, and products to get a review representation for fake reviews detection. Li et al. [29] presented a deep social network detection model for spammers. Yelundur et al. [30] proposed a binary multiobjective method to detect review abuse. Their experimental results show that the model achieves higher accuracy/recall than unsupervised technology. Wang et al. [31] constructed a multiple feature fusion and rolling collaborative training model for fake reviews. Imam et al. [32] considered that the characteristics of fake reviews might change over time, so they proposed a semi-supervision model for the drift problem.

C. PU Learning Algorithm

PU learning algorithm is a semi-supervised binary classification model. Different from the traditional semi-supervised classification model, the training of PU learning can be done with only a small amount of real reviews [33].

Kiryo et al. [34] propose a nonnegative risk estimator for PU learning that solves the overfitting problem caused by introducing deep learning models in the PU learning field. Niu et al. [35] proved that PU and NU learning given infinite U data will almost always improve on pn learning. Zayd et al. [36] propose a general simplified method to solve the overfitting problem of PU risk estimation. Fang et al. [37], [38] studied the learning world problem of open set learning and unsupervised open set domain adaptation (UOSDA) in the PU algorithm.

PU learning learns through a small number of labeled positive samples, achieving better classification results. Therefore, it has wide applications in the field of fake reviews detection. Fusilier et al. [39] divided fake reviews into positive and negative to use PU learning arithmetic focus on deceptive opinions. Li et al. [40] compared multiple groups of experiments, and the final results showed that the PU learning outperformed the supervised learning in fake review detection. Chen et al. [41] effectively detected spammers in Sina Weibo based on the semi-supervised algorithm. He et al. [42] combined PU learning and behavior density to study the detection of fake reviews.

D. Streaming Data

Compared with static and limited data stored in a database or file, streaming data have the following characteristics: the scale is usually massive, unlimited increments, dynamic changes, and timely decision-making [43]. These features are very consistent with the features and requirements of false comment detection.

Liu et al. [44] proposed a recurrent neural network model based on LSTM and LSTM+, which can correct the abnormal data through the anomalous data in the streaming data. Martín et al. [45] propose Kafka-ML, a novel and open-source framework that enables the management of ML/AI pipelines through streaming data. Sun et al. [46]

proposed a data stream cleaning system using edge intelligence. Ning et al. [47] proposed a new high-dimensional online learning method for online high-dimensional regression and classification problems through efficient polar coordinate decomposition. Liang et al. [48] present an anomaly detection aided budget online weighted learning method to identify positive and negative instances from imbalanced data streams. Singh et al. [49] propose a self-adaptive density summarizing incremental natural outlier detection in the data stream with skipping scheme and without skipping technique, which solves the problem of not detecting the memory.

Based on the above-related works, detecting fake reviews has made significant progress. However, fake reviews detection remains a challenge because of deceptive fake reviews' hidden and diverse features. Based on previous research, a detection model for fake reviews is proposed to improve the accuracy of deceptive fake reviews by changing the dataset environment of reviews placed. The different sentiment values of reviews are used to divide the review into two subsets, and we use an improved PU algorithm, which can continuously learn the prediction model from the constantly arriving streaming data. In addition, we continue to detect fake reviews from the unmarked set through the iteration of the PU algorithm and update the training samples in time based on the detection results, so that the model can continue to learn and predict from the fast-arriving streaming data.

III. MODEL PRINCIPLE AND PARAMETERS OF SIPUL

This section introduces the principle of detecting deceptive reviews in detail. It mainly includes the principle of SIPUL in detecting deceptive fake reviews, the criterion of review division, and the calculation of sentiment threshold.

A. Principle of Detecting Deceptive Reviews

Researchers have studied the detection of fake reviews for many years. Dianping website (a Chinese commodity review website) built a system for fake review detection, and they are confident about the system's accuracy, but they do not know the recall [40]. The rest of the reviews may hide many fake reviews that have not been detected. Usually, the fake reviews that are not detected are mostly deceptive. If deceptive reviews can be effectively detected, it will greatly improve the efficiency of detecting fake reviews.

Deceptive reviews are difficult to detect because they are similar to the real ones and the fake features are not obvious in the given product dataset. As we all know, data and features determine the ceiling of machine learning, while models and algorithms only approach the ceiling. Consequently, a binary machine learning classification problem can get better experimental results if significant and effective features can be extracted from the given data. To detect the deceptive fake reviews better, we can explore from the following two points.

- 1) *Feature*: Extracting representative fake review features from a given dataset to distinguish real reviews.
- 2) *Data*: Changing the set environment (dataset) of the fake reviews placed. The fake reviews with hidden features

in the original environment will relatively lose hidden in the new environment.

Most of the traditional research work started from the first point. However, it is challenging to extract significantly useful features from the dataset due to the hidden feature of deceptive reviews. Therefore, this article studies the second point. Our motivation is to change the environment of the fake review placed by dividing the dataset according to different sentiment intensities. Therefore, the original deceptive reviews with the hidden feature will no longer be hidden in the new set environment (subset). The detailed principle is shown in Fig. 2.

Model explanation: In Fig. 2, r_i and f_i denote the review text and review feature. r_1, r_3, r_5, r_7 and r_9 are assumed as real reviews, while r_2, r_4, r_6, r_8 and r_{10} are fake reviews. According to previous work, we know that different review features have different influences on fake reviews. To explain the principle of the model better, it is assumed that the impact of the ten fake review features is the same, so that the review with more fake features is more likely to be fake. Based on the above assumptions, the comparison and analysis of the previous working principle and our working principle are in the following.

Previous working principle: In Fig. 2, reviews contain at most eight fake features and at least one fake feature. Therefore, the reviews with fake features greater than 4 can be classified as negative examples (fake reviews) and less than or equal to 4 are classified as positive examples (real reviews). The results are shown in Fig. 2. The positive examples include r_1, r_3, r_6, r_7 and r_{10} , and the negative examples include r_2, r_4, r_5, r_8 , and r_9 . According to the previous assumptions, we found that r_5, r_6, r_9 , and r_{10} are misclassified. The comparison with the results of this article is shown in Table I.

Our working principle: Under the same assumption, our working principle adds a feature partition layer to the previous working principle. The layer added is designed to improve the detection of deceptive reviews with the hidden feature by dividing the given dataset into two parts (i.e., strong feature matrix and weak feature matrix). In this case, the number of fake features in the strong feature matrix is 5–8 and in the weak feature matrix is 1–4. In the strong feature matrix, the reviews with fake features greater than 6 are divided into negative examples, and the reviews less than or equal to 6 are divided into real reviews. In the weak feature matrix, reviews with fake features greater than 2 are classified as negative examples, and reviews less than or equal to 2 are classified as real reviews. The detection result is shown in Fig. 2. The positive examples include r_1, r_2, r_5, r_7 , and r_9 , and the negative examples include r_3, r_4, r_6, r_8 , and r_{10} . According to the hypothesis, it can be known that r_2 and r_3 are misclassified.

The result analysis of model principles: In Table I, the result showed that r_1, r_4, r_7 , and r_8 are accurately detected in the two model principles. The reason is that r_1 and r_7 contain fewer fake features, and r_4 and r_8 contain more fake features. r_1 and r_7 contain fewer fake features, which is more obvious

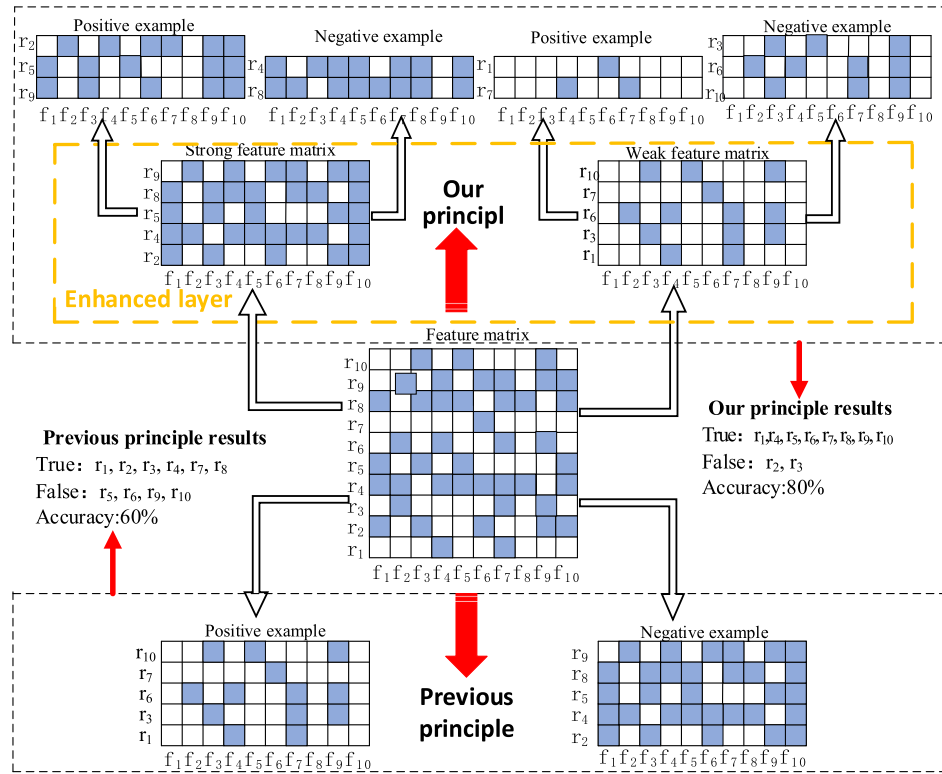


Fig. 2. Comparison of model principles. The yellow dashed line is the dataset division operation (i.e., the calculation of sentiment value) that the work of this article has increased compared with the previous work. $r_i f_i$ to denote the review text and review feature. r_1, r_3, r_5, r_7 , and r_9 are assumed as real reviews, while r_2, r_4, r_6, r_8 , and r_{10} are fake reviews.

TABLE I
COMPARISON RESULTS OF MODEL PRINCIPLES

Review	Real/Fake	Feature number	Previous results	Our results
r_1	Real	2	Real	Real
r_2	Fake	5	Fake	Real (×)
r_3	Real	3	Real	Fake (×)
r_4	Fake	8	Fake	Fake
r_5	Real	5	Fake (×)	Real
r_6	Fake	4	Real (×)	Fake
r_7	Real	1	Real	Real
r_8	Fake	7	Fake	Fake
r_9	Real	6	Fake (×)	Real
r_{10}	Fake	3	Real (×)	Fake
Accuracy			60%	80%

than other reviews. On the contrary, r_4 and r_8 contain more fake features, which is also more obvious than other reviews. These prominent features make these reviews easy to detect correctly.

However, the reviews containing three to six fake features have a higher overall similarity between real reviews and fake reviews. The reviews in this part are intertwined, which is equivalent to hide the fake reviews in the real reviews. It is difficult to detect deceptive reviews with hidden features because of the single feature extracted directly from the given dataset. Our method is to separate the intertwined parts of

reviews as much as possible according to different sentiment values before the model is trained. In dividing the review, most of the fake reviews tended to be classified into the strong sentiment set, while real reviews tended to be classified into the weak sentiment set. This division indirectly changes the set environment of the fake reviews placed, which makes deceptive reviews with hidden features no longer have hidden features in the subset. Therefore, it is easier to detect deceptive fake reviews in the divided subset. The effectiveness of this model is verified through experiments in Section V.

TABLE II
DISTRIBUTION OF THE DATASET

Data type	Num- ber	Text length ≥ 50	Text length ≤ 50	Num- ber
Fake review	3412	2536	876	3412
Real review	9834	1458	8376	9834
Total	13246	3994	9252	13246

B. How to Select Criterion of Review Division

According to the analysis in Section III-A, it is very crucial to choose a representative criterion for review division. In this article, sentiment intensity is selected as the criterion for the classification of the review.

In general, many features may affect fake reviews. We analyzed ten common features based on [50]. The ten features are text length (F1), text complexity (F2), relevance between reviews and products (F3), consistency of reviews and ratings (F4), sentiment intensity (F5), whether to include transitional words (F6), copy text (F7), user reputation (F8), consistency of evaluation (F9), and attached advertising picture (F10). They use the stepwise regression to calculate which feature has a severe influence on fake reviews, and the advantage of the feature is calculated by the odds ratio (OR) [51]. The OR is a ratio index to measure the effect of risk factors in epidemics, and its calculation formula is as follows:

$$OR_j = \frac{P_1/(1 - P_1)}{P_0/(1 - P_0)} \quad (1)$$

where P_1 and P_0 denote the probability of a false attack when the independent variable X_j denotes different values. OR_j denotes the multivariate-adjusted OR. Next, on the dataset collected by Otter et al. [20] (shown in Table II), we used the length of the review as an example to expound the calculation of the OR

$$\text{Text length} \geq 50 \text{ advantage} = \frac{2536/3994}{1458/3394} \approx 1.74 \quad (2)$$

$$\text{Text length} < 50 \text{ advantage} = \frac{876/9252}{8376/9250} \approx 0.10. \quad (3)$$

Therefore, $OR = (1.74/0.10) = 17.4$, and the effect of text length on fake reviews detection is very significant. Based on this principle, the ORs of ten features are calculated, and the result is shown in Table III.

As shown in Table III, we can find that the three largest ORs are 17.4 of the length of the text, 15.6 of whether it contains turning words, and 7.68 of the sentiment intensities.

C. Calculation of Sentiment Value

Considering the influence of different fields and language habits, the definition of strong or weak sentiment is not a definite value. The sentiment threshold needs to be determined according to the field represented by the review and the collection of datasets. In addition, the sentiment value obtained by using different sentiment calculation methods for the same reviews is not necessarily the same. This article adopts our

TABLE III
OR VALUE

Feature	Assignment description	OR
F1	Text length $\geq 50=1$, Text length $< 50=0$	17.4
F2	Complex =1, Uncomplex =0	5.31
F3	Irrelevant =1, Relevant =0	0.34
F4	Inconsistency =1, Consistency=0	3.87
F5	Strong =1, Weak=0	7.68
F6	Not included =1, Included =0	15.6
F7	Yes=1, No=0	2.87
F8	High=1, Low=0	0.87
F9	Inconsistency =1, Consistency =0	7.46
F10	Attached pictures =1, No pictures =0	4.23

existing research results (SC-CMC-KS) as the theoretical algorithm of sentiment computing, and the theory of the method comes from [7]. This article divides the reviews according to the intensity of the sentiment and does not consider the positive or negative. Therefore, the absolute value operation is performed on the calculation result of the algorithm

$$FS = |\lambda \times \text{TScore}(\text{total}) + (1 - \lambda) \times \text{SE}(\text{emotion})| \quad (4)$$

where FS is the sentiment value of the review, TScore(total) is the total sentiment value of sentiment words in the review, SE(emotion) is the sentiment value of the emoticon, and λ is the ratio of text, and the detailed calculation process is referred to [7].

IV. CONSTRUCTION OF THE MODEL SIPUL

This section will introduce the construction of the model SIPUL in detail. It mainly includes the introduction of the PU learning for fake review detection and three steps of the construction process of SIPUL.

A. PU Learning

The PU learning algorithm learns from a small number of positive examples [46]. First, it identifies reliable negative examples with higher credibility from the unlabeled dataset. Then, iterative training classification models until a specific stop condition are reached according to the extracted samples. Generally speaking, there are three types of PU learning algorithms: two-step PU learning, biased PU learning, and incorporation of the class prior. Based on the two PU algorithms, this article controls the iteration of the model and the update of training sample data to deal with the increasing and changing characteristics of review streaming data.

This article uses two-step PU learning. The workflow of the algorithm is shown in Fig. 3.

Fig. 3 shows the process of using the PU learning algorithm to construct a classifier to detect fake reviews. There are only positive and no negative examples given a training set P. The streaming data after sentiment division are put into the unlabeled set U for detection. The U contains both positive

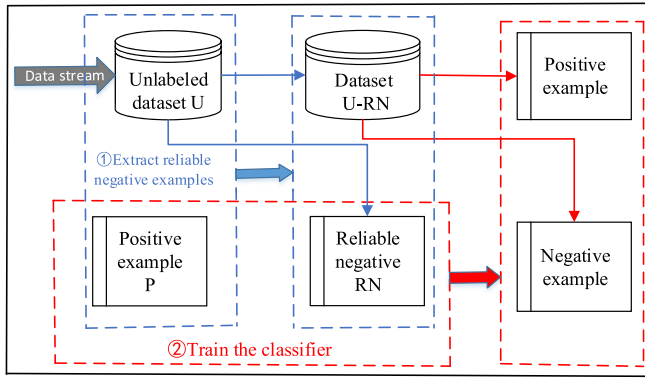


Fig. 3. Construction process of a two-stage PU learning algorithm for streaming data. The blue arrow indicates the extraction of negative examples, and the red arrow indicates model training and classification.

and negative examples. PU learning algorithm mainly includes the following two steps.

- 1) *Extracting reliable negative examples*: According to the positive sample P and unlabeled dataset U , reliable negative samples (RN) are extracted from U . The set U is divided into RN and U -RN.
- 2) *Training the classifier iteratively*: The set P and RN form the initial training set, and a traditional binary classifier (e.g., SVM) is trained iteratively by constantly updating set P and RN. The iteration does not stop until a stop condition is reached.

The main feature of the PU learning algorithm is that it does not require negative training examples. Our goal is to build a binary classifier using P and U for fake review detection that continuously detects reviews in unlabeled reviews.

B. Construction Process of SIPUL

The problem of fake reviews detection is defined in the PU learning algorithm. According to the previous description, a real sample review dataset is required first. The positive and negative examples of initial training need to be obtained before the model trainer and are not needed in subsequent model iterations. Although people cannot effectively detect fake reviews in the current Internet environment, they can collect a small number of real reviews through previous knowledge and heuristic rules. This article shows the strategy for collecting real reviews in Fig. 4.

P represents the positive sample set, and U means the unlabeled reviews. To get a sufficiently reliable positive example set P , we require only samples with sufficient confidence to be put into set P . For the uncertain samples, the traditional way is to put them back into the set U directly, so that they might be picked up again. We introduce a temporary set I , which is used to temporarily store the uncertain samples, to ensure that every time the samples extracted from U are not extracted.

Based on the datasets P and U , this article presents a fake review detection model based on SIPUL, and the PU learning mainly includes three steps.

- Step1. Extracting reliable negative examples.
- Step2. Training the model iteratively.

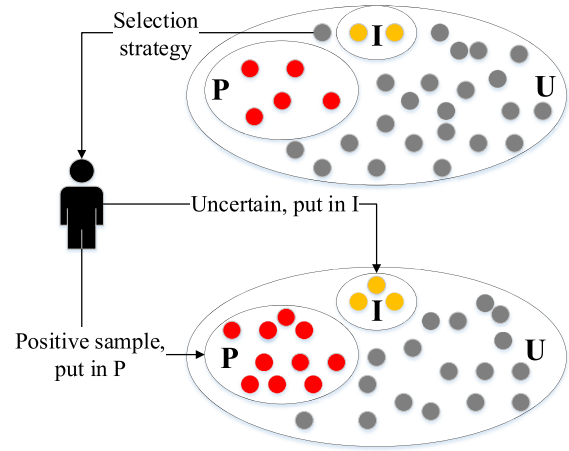


Fig. 4. Strategy for collecting positive examples. U , P , and I represent the unlabeled text, positive example, and temporary collection, respectively.

Step3. Updating training samples and historical record points.

The following sections will describe each step in detail.

1) *Extracting Reliable Negative Examples*: First, the SIPUL algorithm needs to extract some negative examples from the unlabeled dataset. In previous work, there are mainly four algorithms to achieve this step: Spy Technique [52], Rocchio [53], NB [54], and 1-DNF [55] technology. Experiments show that the combined performance of Spy and SVM is the best. Therefore, this article uses Spy to extract reliable negative examples. The algorithm of the Spy Technique is shown in Algorithm 1.

Algorithm 1 Extracting Reliable Negative Examples

Input: Example set P and U , sampling rate s

Output: Example set RN

- 1: Set $RN = \emptyset$, $s = P \times S\%$
- 2: Set $P_s = P - S$, $U_s = U \cup S$, the label of P_s to 1, the label of U_s to -1
- 3: According to P_s and U_s to train a binary classifier g
- 4: Using g to classify example in set U
- 5: Determine the threshold ϕ
- 6: **for** $a \in U$ **do**
- 7: if $P_r(1|a) \leq \phi$
- 8: $RN = RN \cup a$
- 9: **end for**
- 10: Output Example set RN

Algorithm 1 describes the main process of extracting reliable negative examples.

- 1) Steps 1–3: Select some samples from the set P randomly and put them into the set S . Then, put the samples from set S into U as Spy samples. At this time, the sample set becomes $P-S$ and $U + S$. Generally, the number of subsets S divided from P is 15%.
- 2) Steps 4–6: $P-S$ and $U + S$ are taken as positive and negative samples, respectively, and then combined with the iterative EM algorithm for classification. All unlabeled samples are trained as negative examples in the initialization process to train the classifier.
- 3) Steps 7–10: Taking the minimum value of Spy sample distribution as the threshold, all samples in U below this threshold are considered as RN.

Algorithm 2 The Iterative Training of the Model

Input: Example set P and U , the negative example set RN
Output: model

- 1: Set model-list= $\emptyset$, $D = U - RN$
- 2: Set the label of P to 1, and the label of RN to -1
- 3: $i = 1$
- 4: **Loop**
- 5: Train the SVM classifier S_i according to P and RN
- 6: Save S_i to model-list
- 7: Use S_i to classify D
- 8: Set the negative review into M in D
- 9: **if** $M = \emptyset$
- 10: exit loop
- 11: **else**
- 12: $D = D - M$
- 13: $RN = RN \cup M$
- 14: $i++$
- 15: **Output** model

2) *Training the Model:* According to the discussion in Section IV-A and IV-B, positive samples and negative samples have been obtained. Then, we need to train a classifier based on P and RN . The commonly used binary classification models are SVM [56] and EM [57]. In this article, SVM is selected as the training model. We aim to train an SVM binary classifier through set P and set RN . The specific algorithm process is shown in Algorithm 2.

Algorithm 2 design description is given as follows.

- 1) Steps (1–6): The set P and set RN to form the initial training set, which is used to build an initial SVM classifier S_i .
- 2) Steps (7–8): The reviews in D are classified by using the trained classifier S_i , and the negative examples are stored in the set M .
- 3) Steps (9–10): This is a judgment statement. If the set M is empty, it means that no negative examples are found in D . In other words, all negative examples have been classified, and the algorithm ends. On the contrary, if the M set is not empty, continue to Line 11.
- 4) Steps (11–13): The identified negative sample is removed from set D , and the negative sample set W is merged into the reliable negative sample set RN . After reassigning D and RN , return to step 4.
- 5) Steps (4–13): This is a process of iteratively training the classifier. The classifier after each training cycle is stored in the model.

3) *Updating Training Samples and Historical Record Points:* The set P and set RN form the initial training set, by constantly updating set P and set RN , and the SVM fake review detector is trained. However, compared to static and limited data stored in a database or file, streaming data have the characteristics of a huge processing scale, unlimited data increment, and dynamic changes. The design principle is shown in Fig. 5.

As shown in Fig. 5, product reviews continue to enter the model. When the data arrive, the model trains and predicts based on the sample data in time and records the training sample data and the trained model at this time. The predicted result is treated as sample data for the next model training. As the sample data continue to increase, the past data in

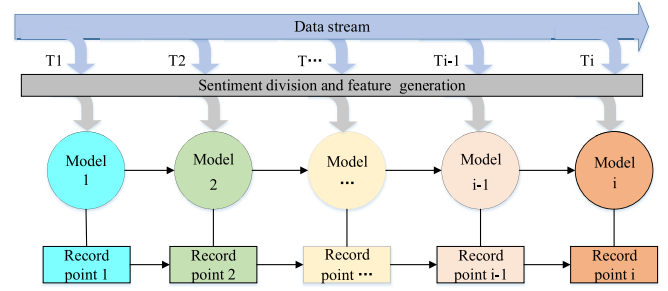


Fig. 5. Updating training samples and historical record points. T_i represents different times of receiving streaming data.

Algorithm 3 Updating Training Samples and Historical Record Points

Input: Product Reviews, Sentiment threshold T
Output: Classification result

- 1: Index= i , Current-model=model(i)
- 2: Product review preprocessing
- 3: Calculating the sentiment value FS of the review
- 4: **if** $FS \geq T$
- 5: Enter the strong sentiment set
- 6: **else**
- 7: Enter the weak sentiment set
- 8: Use model(i) to detect fake reviews in the set
- 9: The result is added to the training sample
- 10: Record the current data, and delete the old data.
- 11: Train the model($i+1$) based on sample data
- 12: **Output:** Classification result

the sample are deleted according to the model capacity. In our model, the last three batches of data are retained. The persistence task will continuously delete old data, keeping the database at a manageable size. We currently support file-based implementation (file streaming), where we simulate data in a streaming manner. We plan to support network-based streaming in the near future. The process of updating training samples and historical record points is shown in Algorithm 3.

Algorithm 3 as a whole is an iterative loop operation:

- 1) Step 1 is the current model cycle record.
- 2) Steps 2–7 are used to calculate the sentiment value of the review and the sentiment division of the review.
- 3) Sept 8 is used to complete the review detection operation.
- 4) Steps 9 and 10 are used to update training samples and record points.
- 5) Step10 is used for iterative training based on updated samples that are used to judge the current model error and the existing minimum error and judge whether to update the final model.

V. EXPERIMENT

A. Experimental Datasets

The dataset reviews include two types: real reviews and fake reviews. To evaluate the effectiveness of our model, we conducted experiments with four public datasets. The information of the experimental datasets is shown in Table IV.

Ott et al. [13] contain deceptive reviews of hotels from crowdsourcing platforms and the same hotel in Chicago by TripAdvisor. YelpChi [58] contains real business reviews of

TABLE IV
EXPERIMENTAL DATASETS

Dataset	Data type	Review topic	Number
Ott	Fake	Hotel	400
	Real		400
YelpZIP	Fake	Restaurant	80466
	Real		528132
YelpChi	Fake	Restaurant and hotel	8919
	Real		58476

restaurants and hotels from Chicago on the Yelp website. YelpZIP [59] contains only restaurants reviews from different regions of Yelp.

In this article, we only use the public dataset stored in files to imitate streaming data's increasing and changing characteristics. We plan to support network-based streaming soon.

B. Experimental Method

Product reviews often appear in the form of unlimited streaming data. This article uses the existing public datasets to simulate the characteristics of streaming data. We currently support file-based implementation (file streaming), and we plan to support network-based streaming soon.

In the experiment, P, RN, and U represent the positive examples, negative examples, and unlabeled samples, respectively. To analyze the performance of the model, different experimental indicators are used to analyze the performance of the model in this article on different datasets. We design three basic experiments (Experiments 1–3) and an experiment that simulates the characteristics of streaming data (Experiment 4).

1) Analyzing the Influence of Different Combinations of Two-Part PU Algorithms:

Step 1: select Spy, Rocchio, NB, and 1-DNF technology. Step 2: select SVM, EM, and NB. We designed eight groups of different combinations of two technologies (as shown in Table VI) and conducted a comparative experiment on “YelpZIP”; 30% of the data are randomly selected as test documents for each dataset. The rest are used to create training sets. γ percentage of the data is first selected as the positive set, and we range γ from 1% to 10%.

2) Analyzing the Influence of Different Sentiment Values on the Experiment:

This experiment is mainly used to analyze the impact of different sentiment values on the experiment's performance and find the sentiment threshold for dividing comments.

3) Analyzing the Influence of Different Review Division Indexes on the Model:

We designed four kinds of feature combinations (as shown in Table V) and conducted a comparative experiment on three datasets: “Ott,” “YelpZIP,” and “YelpChi.” The combination of PU and other representative fake reviews shows that it is reasonable to choose emotional intensity as the classification of comments.

TABLE V
COMBINATION DESIGN OF DIFFERENT FEATURES

Number	Feature combination
C1	PU(Spy+SVM)
C2	PU(Spy+SVM) +Text length
C3	PU(Spy+SVM) + Sentiment intensity
C4	PU(Spy+SVM) + Transitional words

Combination 1(C1) is the template, regardless of the division of reviews. Combination 2–4 (C2–C4), respectively, indicate that text length, sentiment intensity, and whether there are transitional words are used as the basis for the division of reviews.

At the end of experiment 3, in order to prove the effectiveness of the SIPUL in detecting fake reviews better, we compared it with several advanced methods.

4) Experiments to Simulate the Characteristics of Streaming Data:

One of the characteristics of streaming data is a large amount of data, so we do simulation experiments on the two datasets: “YelpZIP” and “YelpChi.” For each dataset, 8% of the positive example documents are randomly selected as the positive set. The rest and all negative example documents are taken as unlabeled data. The unlabeled data are divided into ten parts to simulate the change of streaming data.

To evaluate the experimental results, accuracy (Acc), area under the ROC (receiver operating characteristic) curve (AUC), and *F-score* are applied for the performance evaluation measures.

C. Experimental Results and Analysis

We have done the following experiments to verify the performance of the model in this article.

1) Experiment 1: Analyzing the Influence of Different Combinations of Two-Part PU Algorithms:

According to Table VI, we can make the following analysis.

1) *With the combination of Spy and SVM, the average F value performs best:* The first step is to adopt 1-DNF, which is very poor performance. The reason is that 1-DNF can go wrong without many positive documents. On the contrary, training Spy does not require a lot of data, so the performance is better. The other six combinations have relatively good performance, but Spy and SVM have the best F-score performance. The reason is that SVM is a stronger classifier than NB, and EM uses NB.

2) *The average F-score increases with R but gradually becomes flat when it reaches a certain value:* When the positive set is very small, the number of spies put into the unlabeled set is also very small, so the generated RN set is unreliable. However, when the number of positive sets reaches a certain level ($\gamma = 8\%$), it can be ensured that the generated RN set is reliable, and with the increase of γ , the performance of F-score will not increase.

TABLE VI
AVERAGE F-SCORES ON YELPZIP

Step1	1-DNF	1-DNF	Spy	Spy	Rocchio	Rocchio	NB	NB
Step2	EM	SVM	SVM	EM	EM	SVM	EM	SVM
$\gamma=1\%$	0.124	0.247	0.105	0.091	0.357	0.284	0.346	0.181
$\gamma=2\%$	0.122	0.301	0.247	0.224	0.378	0.419	0.441	0.238
$\gamma=3\%$	0.125	0.328	0.485	0.398	0.443	0.446	0.489	0.474
$\gamma=4\%$	0.123	0.298	0.665	0.512	0.569	0.590	0.568	0.556
$\gamma=5\%$	0.119	0.266	0.731	0.654	0.570	0.648	0.665	0.588
$\gamma=6\%$	0.124	0.392	0.762	0.682	0.623	0.696	0.705	0.665
$\gamma=7\%$	0.121	0.312	0.794	0.716	0.654	0.722	0.702	0.701
$\gamma=8\%$	0.124	0.378	0.802	0.743	0.665	0.729	0.718	0.741
$\gamma=9\%$	0.126	0.287	0.802	0.744	0.667	0.731	0.716	0.743
$\gamma=10\%$	0.125	0.276	0.803	0.743	0.665	0.737	0.717	0.744

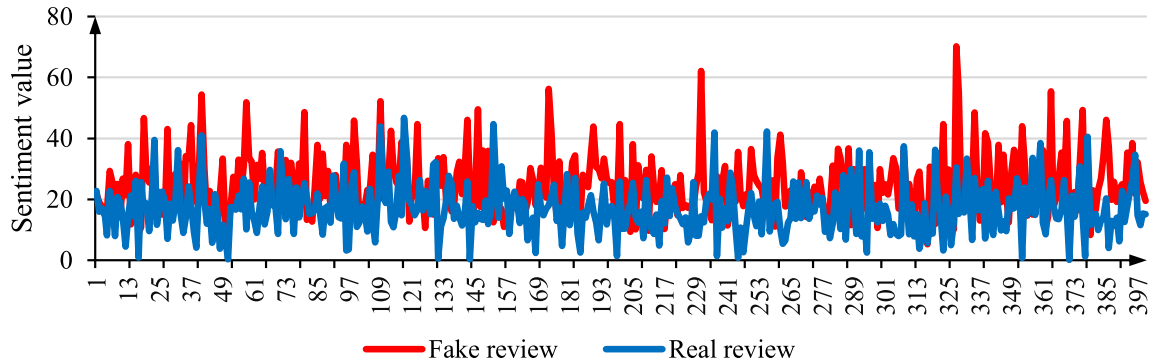


Fig. 6. Sentiment value of different reviews. The red curve representing the fake review is above the blue curve representing the true review overall.

2) Experiment 2: Analyzing the Influence of Different Sentiment Values on the Experiment:

According to the sentiment calculation method provided in Section III-C, the sentiment values of all reviews were calculated, respectively. Because of the large scale of the dataset, we randomly selected 400 real reviews and fake reviews, and the results are shown in Fig. 6.

From the sentiment calculation results of reviews in Fig. 6, it can be found that most of the sentiment value mainly fluctuates between 15 and 25 and mainly concentrates 20. Therefore, we select sentiment values of 15, 20, and 25 as sentiment thresholds to divide the review.

We designed three groups of comparative experiments on three datasets. For each dataset, we set $\gamma = 8\%$, 30% of the data are as test set, and the rest (70%) are used to create training. The average results of the experiment are shown in Table VII and Fig. 7.

According to the experimental results of the three thresholds, we summarize the following two important results.

- 1) *With the increase of the threshold, the experimental index values are roughly in an inverted U-shaped distribution:* We can find from Fig. 7 that when the sentiment threshold is set to 20, the accuracy, precision, recall, and F1 are higher than those of 15 or 25. The overall trend is that the selected index value first increases and then decreases with the threshold value increase. The reason

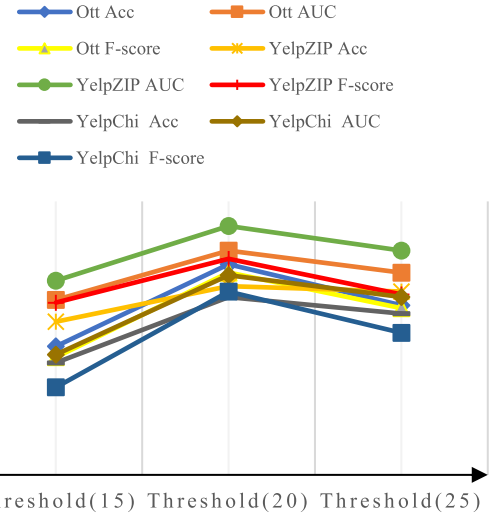


Fig. 7. Influence of sentiment value on experimental performance. Ott ACC, Ott AUC, and Ott F-score represent the evaluation index values of ACC, AUC, and F-score on the dataset Ott. YelpZIP and YelpChi are the same as above.

is that the threshold is too large or too small to separate the real reviews and fake reviews.

- 2) *The sentiment intensity threshold is selected where the most concentrated sentiment value distribution:* The sentiment value of most reviews is mainly concentrated on

TABLE VII
INFLUENCE OF SENTIMENT VALUE ON EXPERIMENTAL PERFORMANCE

threshold	Ott dataset			YelpZIP dataset			YelpChi dataset		
	Acc	AUC	F-score	Acc	AUC	F-score	Acc	AUC	F-score
15	0.847	0.864	0.843	0.856	0.871	0.863	0.841	0.844	0.832
20	0.877	0.882	0.874	0.869	0.891	0.879	0.865	0.873	0.867
25	0.862	0.874	0.861	0.867	0.882	0.866	0.859	0.865	0.852

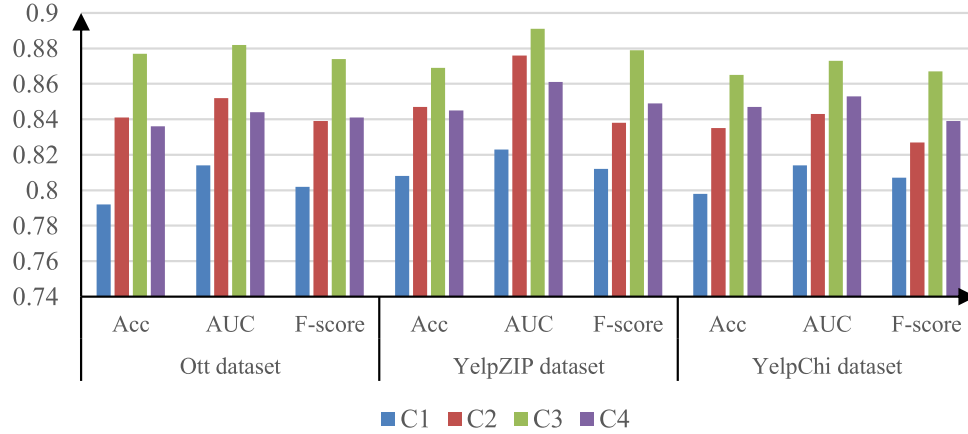


Fig. 8. Influence of different review division indicators on the model. C1 is the template, regardless of the division of reviews. C2–C4, respectively, indicates that text length, sentiment intensity, and whether there are transitional words are used as the basis for the division of reviews.

the upper and lower 20. When the sentiment is divided by the sentiment threshold of 20, the intertwined reviews are separated as much as possible. Compared with the original set environment, the subset environment of fake reviews is changed dramatically. The features of fake reviews are shown as much as possible in the subset.

3) Experiment 3: Analyzing the Influence of Different Review Division Indexes on the Model:

In this experiment, the PU algorithm uses a combination of Spy and SVM, choosing $\gamma = 8\%$. According to Fig. 8, the following two conclusions can be drawn.

- 1) *The classification of reviews can effectively improve the performance of the model in detecting fake reviews:* Compared with the other three groups (C2–C4), the performance of C1 is poor. The main reason is that C1 uses the most primitive dataset. The deceptive reviews that originally have hidden features are very similar to real reviews and are difficult to identify. After C2–C4 uses different features to divide the reviews, the reviews are divided into strong and weak feature sets, and then, they are trained and detected by different models.
- 2) *Sentiment intensity can achieve a better division of reviews:* According to the knowledge, we know that the length of the text and whether the text contains transitional words are essential features that may affect sentiment intensity. However, the text with strong sentiment intensity is not necessarily long or contains transitional words. That is to say, text length and whether there are transitional words can be represented by sentiment intensity, but the strength of sentiment intensity cannot

be directly represented by text length or whether there are transitional words.

To prove the effectiveness of the SIPUL in detecting fake reviews better, we compared it with several advanced methods, and the results are shown in Table VIII.

On the Ott dataset, our method does not perform very well. The reason is that the number of Ott dataset is small, and less data are used for training after sentiment division. However, it performs well on the YelpZIP and YelpChi datasets, but the time cost is greater than other methods. The reason is that we added the sentiment division step before the review detection. Although the accuracy is improved, the training time is increased.

4) Experiment 4: Experiments to Simulate the Characteristics of Streaming Data:

From Fig. 9, it can be found that when the amount of data is increased in batches over time, the model performance improves in the first three times but then begins to decline slowly. The reason is that the model learned more features when adding data for the first three times. However, when the amount of data increases, the model does not learn more new features. Instead, the results of each detection (uncertain and 100% correct) are added to the training sample. The unreliable reviews in the training sample are increasing, resulting in the decline of model performance.

To further prove the effectiveness of our experiment, after simulating the stream data, we only take the data saved in the first three recording points for experimental simulation. As shown in Fig. 10, it can be found that only the latest three recording points are used to train the model each time, and the performance of the model gradually stabilizes after the third

TABLE VIII
EXPERIMENTAL RESULTS OF DIFFERENT FAKE REVIEW DETECTION MODELS IN THE DATASETS: OTT, YELPZIP, AND YELPCHI

Method	Ott dataset				YelpZIP dataset				YelpChi dataset			
	Acc	AUC	F-score	Test time(s)	Acc	AUC	F-score	Test time(s)	Acc	AUC	F-score	Test time(s)
SAE [23]	0.811	0.775	0.796	0.023	0.796	0.793	0.812	532.619	0.788	0.804	0.810	64.734
SPR2EP [21]	0.824	0.833	0.845	0.001	0.831	0.843	0.841	237.982	0.843	0.839	0.837	35.613
ABNN [10]	0.803	0.823	0.814	0.057	0.822	0.817	0.816	394.293	0.801	0.804	0.798	43.753
HFAN [28]	0.872	0.884	0.859	0.012	0.853	0.876	0.865	369.451	0.844	0.864	0.853	35.792
DFNN [11]	0.855	0.892	0.833	0.014	0.798	0.836	0.813	456.242	0.803	0.829	0.830	55.634
SIPUL	0.862	0.874	0.861	0.002	0.867	0.891	0.866	673.668	0.859	0.865	0.852	73.891

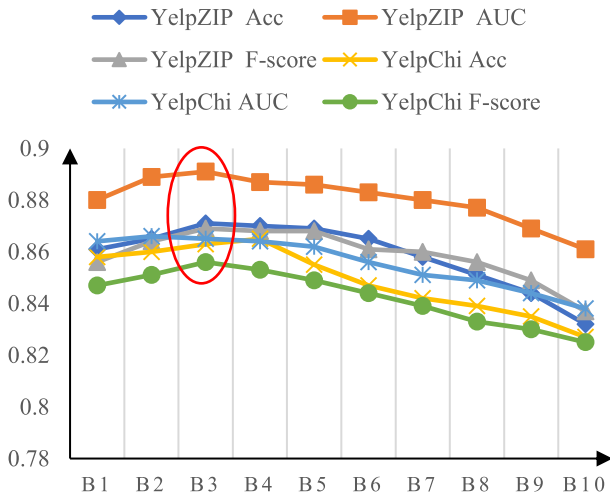


Fig. 9. Experiments simulating streaming data on datasets: YelpZIP and YelpChi. B1–B10 represent the ten parts of the dataset and the amount of data added to the model over time.

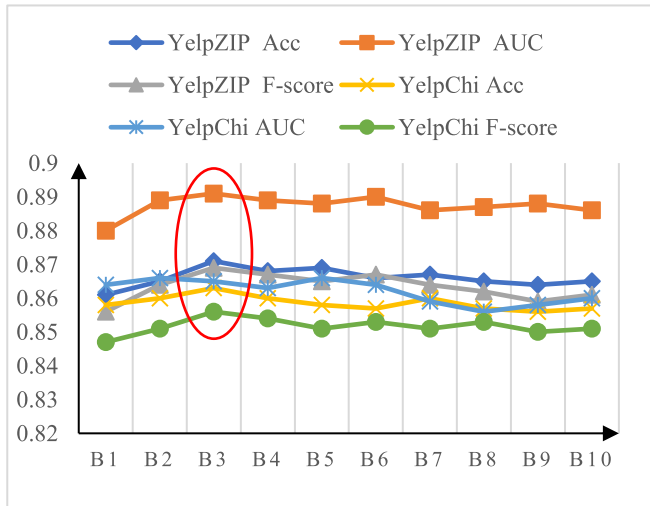


Fig. 10. Experiments simulating streaming data on datasets: YelpZIP and YelpChi.

simulation data. However, compared to the best case, there is still a slow decline. It shows that the test results cannot be guaranteed to be completely correct, and these uncertain data still have a certain impact on the model.

In the comparison of the results of different experimental indicators, although there are slight fluctuations in the

experiments on different datasets, the models proposed in this article all show good performance in the comparison of advanced models, and it also shows good performance of fake reviews detection in the simulated streaming data experiment.

VI. CONCLUSION

To support consumers' understanding of the true quality, performance, and user's evaluation of the product, a new detection model for fake reviews based on SIPUL is proposed, where natural language processing and PU learning algorithm techniques are applied to fake reviews detection. The three contributions are summarized as follows.

- 1) *Selecting the sentiment intensity as the representative review feature:* It is verified that sentiment intensity better represents many fake features. Therefore, in the process of review division, more fake reviews are obtained in the strong sentiment set, and more real reviews are obtained in the weak sentiment set.
- 2) *A method to detect deceptive fake reviews is proposed:* This article creatively proposes that by changing the dataset environment where the fake reviews are located, the fake reviews with hidden features will lose their hidden in the new set environment, so that they can be better detected.
- 3) *Fake review detection model SIPUL is successfully constructed:* Experiments show that the model is effective in detecting fake reviews. In addition, by controlling the iteration of the model and the update of training data, the model can continuously learn and predict from the continuously arriving streaming data. The characteristics of simulated streaming data preliminarily prove the effectiveness of this method.

The model currently only supports file-based implementation (file streaming), and it needs to update the training data according to the record constantly; otherwise, the performance will be affected. We plan to support network-based streaming soon.

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